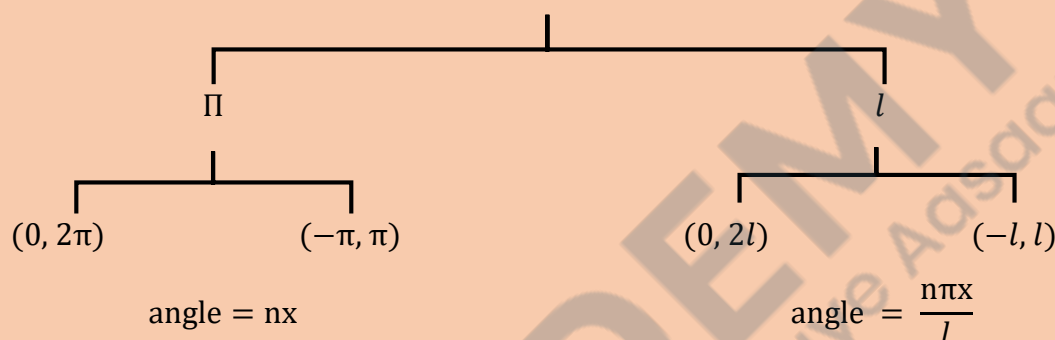


FOURIER SERIES

TYPES

- 1) Examples on $(0, 2\pi) / (0, 2l)$
- 2) Examples on $(-\pi, \pi) / (-l, l)$
- 3) Half Range (cosine / sine) series

INTERVAL



Fourier series Formula

I. For 'l'

OR

1) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$	$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$
2) $a_0 = \frac{1}{l} \int f(x) \cdot dx$	$a_0 = \frac{1}{2l} \int f(x) \cdot dx$
3) $a_n = \frac{1}{l} \int f(x) \cos\left(\frac{n\pi x}{l}\right) \cdot dx$	$a_n = \frac{1}{l} \int f(x) \cos\left(\frac{n\pi x}{l}\right) \cdot dx$
4) $b_n = \frac{1}{l} \int f(x) \sin\left(\frac{n\pi x}{l}\right) \cdot dx$	$b_n = \frac{1}{l} \int f(x) \sin\left(\frac{n\pi x}{l}\right) \cdot dx$

II. For 'π'

OR

1) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$	$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$
2) $a_0 = \frac{1}{\pi} \int f(x) \cdot dx$	$a_0 = \frac{1}{2\pi} \int f(x) \cdot dx$
3) $a_n = \frac{1}{\pi} \int f(x) \cdot \cos nx \cdot dx$	$a_n = \frac{1}{\pi} \int f(x) \cdot \cos nx \cdot dx$
4) $b_n = \frac{1}{\pi} \int f(x) \cdot \sin nx \cdot dx$	$b_n = \frac{1}{\pi} \int f(x) \cdot \sin nx \cdot dx$

General Store

1) $\sin 0 = \sin n\pi = 0$

2) $\cos 0 = \cos 2\pi n = (-1)^{2n} = 1$

3) $\cos n\pi = (-1)^n$

4) $\frac{d}{dx} \text{const} = 0$	$\int \text{const} \cdot dx = \text{const} \cdot x$
$\frac{d}{dx} (x^n) = n \cdot x^{n-1}$	$\int x^n \cdot dx = \frac{x^{n+1}}{n+1}$

5) $ds = c$	$\int s = -c$
$dc = -s$	$\int c = s$

6) $\int (u \cdot v) dx = + (u)(I) - (D)(I) + (D)(I) - \dots$

Master Key

$\uparrow \pi = nx$ Angle $\downarrow l = \frac{n\pi x}{l}$	$\frac{d}{dx} \text{const} = 0$	$\int \text{const} \cdot dx = \text{const} \cdot x$	$\int s = -c$	$\sin 0 = \sin n\pi = 0$
	$\frac{d}{dx} (x^n) = n \cdot x^{n-1}$	$\int x^n \cdot dx = \frac{x^{n+1}}{n+1}$	$\int c = s$	$\cos 0 = \cos 2\pi n = (-1)^{2n} = 1$
				$\cos n\pi = (-1)^n$

Half Range Series

1. Half range cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(\quad)] \quad \text{or} \quad f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(\quad)$$

For π interval	For l interval	For π interval	For l interval
$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$	$a_0 = \frac{2}{l} \int_0^l f(x) dx$	$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx$	$a_0 = \frac{1}{l} \int_0^l f(x) dx$
$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$	$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$	$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$	$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$

2. Half range sine series

$$f(x) = \sum_{n=1}^{\infty} [b_n \sin(\quad)]$$

For π interval

$$b_n = \frac{2}{\pi} \int f(x) \sin nx dx$$

For l interval

$$b_n = \frac{2}{l} \int f(x) \sin \frac{n\pi x}{l} dx$$

Parseval's identity

$$\int f^2(x) dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

RK **RKD** **EM**
Education Banaye Aasaan